Hamza Ibrahim El-Sayed

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PROFESSIONAL SUMMARY

Third-year Computer Science and Artificial Intelligence student with hands-on data engineering experience in Python, SQL, data pipelines, and data warehousing. Building machine learning fundamentals through training and projects, including a CNN image classifier and ensemble classification models. Seeking an AI, Machine Learning, or Data Engineering internship.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++
- **Machine Learning & AI:** Supervised Learning, Unsupervised Learning, Model Evaluation, Natural Language Processing (NLP)
- **Deep Learning & Computer Vision:** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Computer Vision
- **Python Libraries:** NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, TensorFlow/Keras, XGBoost
- **Data Engineering & Databases:** SQL, Database Management, Data Pipelines, Data Warehousing, Big Data Processing
- **Cloud & Visualization:** Microsoft Azure, Power BI
- **Other Skills:** Object-Oriented Programming (OOP), Problem Solving, Prompt Engineering, HTML/CSS/JavaScript (basic)

EXPERIENCE

AI and Machine Learning Trainee, Route Academy *Jul 2026 – Present*

- Training in supervised and unsupervised learning, machine learning algorithms, and model evaluation techniques
- Studying deep learning, artificial neural networks (ANNs), computer vision with CNNs, and natural language processing (NLP)
- Practicing model implementation in Python with NumPy, Pandas, scikit-learn, and TensorFlow/Keras

Data Engineer, Digital Egypt Pioneers Initiative (DEPI) *Nov 2025 – Jul 2026*

- Applied prompt engineering and studied applied AI techniques for AI-driven data engineering workflows
- Performed SQL-based database management and query optimization tasks
- Built and maintained data pipelines for structured and unstructured data processing
- Practiced big data processing concepts and completed training aligned with the Microsoft Azure Data Engineer Associate track

EDUCATION

Helwan University – B.Sc. in Computer Science and Artificial Intelligence *2024 – 2028*

Faculty of Computer Science and Artificial Intelligence | Cairo, Egypt | Grade: Very Good

PROJECTS

CNN Image Classification (Vegetable Image Dataset) | *Python, TensorFlow/Keras, NumPy, Pandas, Matplotlib*

- Prepared 15-class vegetable image data (15,000 train, 3,000 validation, 3,000 test) with ImageDataGenerator (resize, rescale)
- Built a Sequential CNN (3 Conv2D/MaxPooling blocks, 3 dense layers); trained 5 epochs with Adam and cross-entropy loss
- Evaluated accuracy and loss on train, validation, and test sets: 92.8% test accuracy, 93.2% validation accuracy

Ensemble Learning for Dry Bean Classification | *Python, scikit-learn, XGBoost, Pandas, NumPy, Seaborn*

- Cleaned the Dry Bean dataset (13,611 samples, 16 features, 7 classes): removed 68 duplicates, clipped outliers (IQR, Z-score), and applied a stratified 80/20 split, scaling, and label encoding
- Compared decision trees (default and GridSearchCV-tuned), bagging (SVC and decision tree), Random Forest, Gradient Boosting, and XGBoost; tuned n_estimators, max_features, learning rate, and subsample
- Evaluated with classification reports and confusion matrices; ensembles reached 91–92% test accuracy vs. 90% for a single tree

Student Churn Prediction (Data Engineering) – Python Scripts, Database, Data Warehouse, Automation, Power BI
Smart Urban Parking System – Led analysis, design, and implementation of a system architecture project
E-Commerce Clothing Store – Built a responsive front-end website using
HTML, CSS, and JavaScript
Project Management System – Designed and developed an object-oriented desktop application in Java
Government Agency Network Simulation – Designed and configured a multi-router WAN in Cisco Packet Tracer, including NAT/PAT and connectivity troubleshooting

CERTIFICATIONS

- Machine Learning Specialization – Deeplearning.AI
- IBM AI Developer Professional Certificate – IBM
- Data Analysis using Python – DataCamp
- AI and Machine Learning – Sprints x Microsoft Summer Camp

LANGUAGES

- English – B2 | French – Basic